From the given equations of skew lines, we observe that L_1 is parallel to the vector $\vec{b}$ and L_2 is parallel to the vector $\vec{d}$. So, $\vec{b}\times\vec{d}$ is perpendicular to the lines L_1 and L_2.

Let SD be the line segment perpendicular to both the lines L_1 and L_2. Then the vector $\overrightarrow{SD}$ is perpendicular to the vectors $\vec{b}$ and $\vec{d}$ and therefore it is parallel to the vector $\vec{b}\times\vec{d}$.

So, $\dfrac{\vec{b}\times\vec{d}}{|\vec{b}\times\vec{d}|}$ is a unit vector in the direction of $\overrightarrow{SD}$. Then, the shortest distance $|\overrightarrow{SD}|$ is the absolute value of the projection of $\overrightarrow{AC}$ on $\overrightarrow{SD}$. That is,

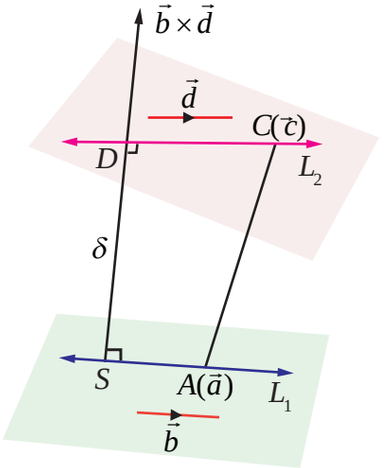


Fig. 6.22

$$\delta = |\overrightarrow{SD}| = |\overrightarrow{AC}| \text{ . (Unit vector in the direction of } \overrightarrow{SD}) | = \left| (\vec{c}-\vec{a})\cdot\frac{\vec{b}\times\vec{d}}{|\vec{b}\times\vec{d}|} \right|$$

$$\delta = \frac{|(\vec{c}-\vec{a})\cdot(\vec{b}\times\vec{d})|}{|\vec{b}\times\vec{d}|}, \text{ where } |\vec{b}\times\vec{d}| \neq 0 .$$

Remark

(i) It follows from theorem (6.14) that two straight lines $\vec{r}=\vec{a}+s\vec{b}$ and $\vec{r}=\vec{c}+t\vec{d}$ intersect each other (that is, coplanar) if $(\vec{c}-\vec{a})\cdot(\vec{b}\times\vec{d})=0$.

(2) If two lines $\dfrac{x-x_1}{b_1}=\dfrac{y-y_1}{b_2}=\dfrac{z-z_1}{b_3}$ and $\dfrac{x-x_2}{d_1}=\dfrac{y-y_2}{d_2}=\dfrac{z-z_2}{d_3}$ intersect each other (that is, coplanar), then we have

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$$

Example 6.34

Find the parametric form of vector equation of a straight line passing through the point of intersection of the straight lines $\vec{r}=(\hat{i}+3\hat{j}-\hat{k})+t(2\hat{i}+3\hat{j}+2\hat{k})$ and $\dfrac{x-2}{1}=\dfrac{y-4}{2}=\dfrac{z+3}{4}$, and perpendicular to both straight lines.

Solution

The Cartesian equations of the straight line $\vec{r}=(\hat{i}+3\hat{j}-\hat{k})+t(2\hat{i}+3\hat{j}+2\hat{k})$ is

$$\frac{x-1}{2}=\frac{y-3}{3}=\frac{z+1}{2}=s \text{ (say)}$$

Then any point on this line is of the form $(2s+1,\ 3s+3,\ 2s-1)$ ... (1)

The Cartesian equation of the second line is $\dfrac{x-2}{1}=\dfrac{y-4}{2}=\dfrac{z+3}{4}=t$ (say)

Then any point on this line is of the form $(t+2,\ 2t+4,\ 4t-3)$ ... (2)

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